

Uncertainty-aware Power System Planning via Gradient Descent

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PowerUp 2026

September 11

University of Colorado, Boulder



PowerUp 2026



Why this problem matters

● Day-ahead Forecasts are imperfect

Wind and solar availability are forecast-driven and non-dispatchable.



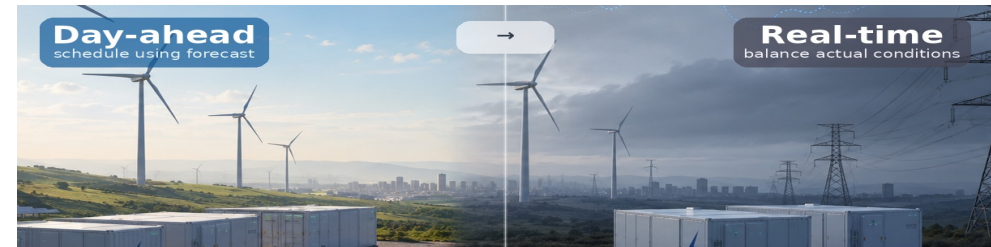
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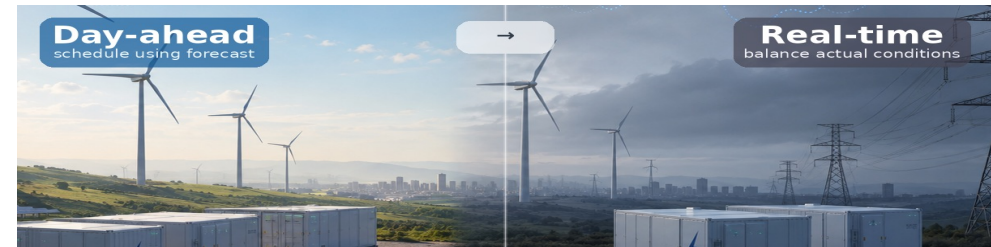
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- **Day-ahead Forecasts are imperfect**
Wind and solar availability are forecast-driven and non-dispatchable.
- **Operations are sequential**
Day-ahead schedules are fixed before real-time uncertainty is realized.
- **Investment is long-lived**
Underinvesting in resources that help balance forecast errors can lead to higher operating costs.



Why this problem matters

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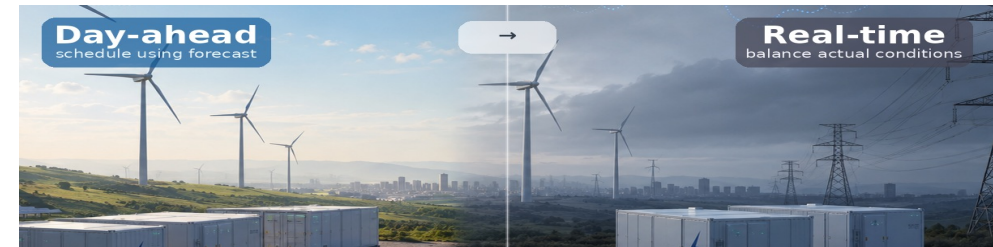
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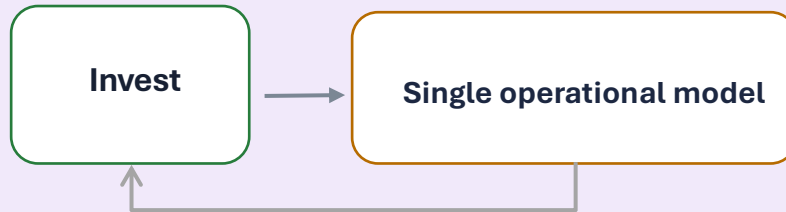


Question: how should capacity expansion change when day-ahead forecasts are wrong?

What standard expansion planning (uncertainty-agnostic model) misses

Ignoring forecast errors hides real-time costs.

UNCERTAINTY-AGNOSTIC BENCHMARK

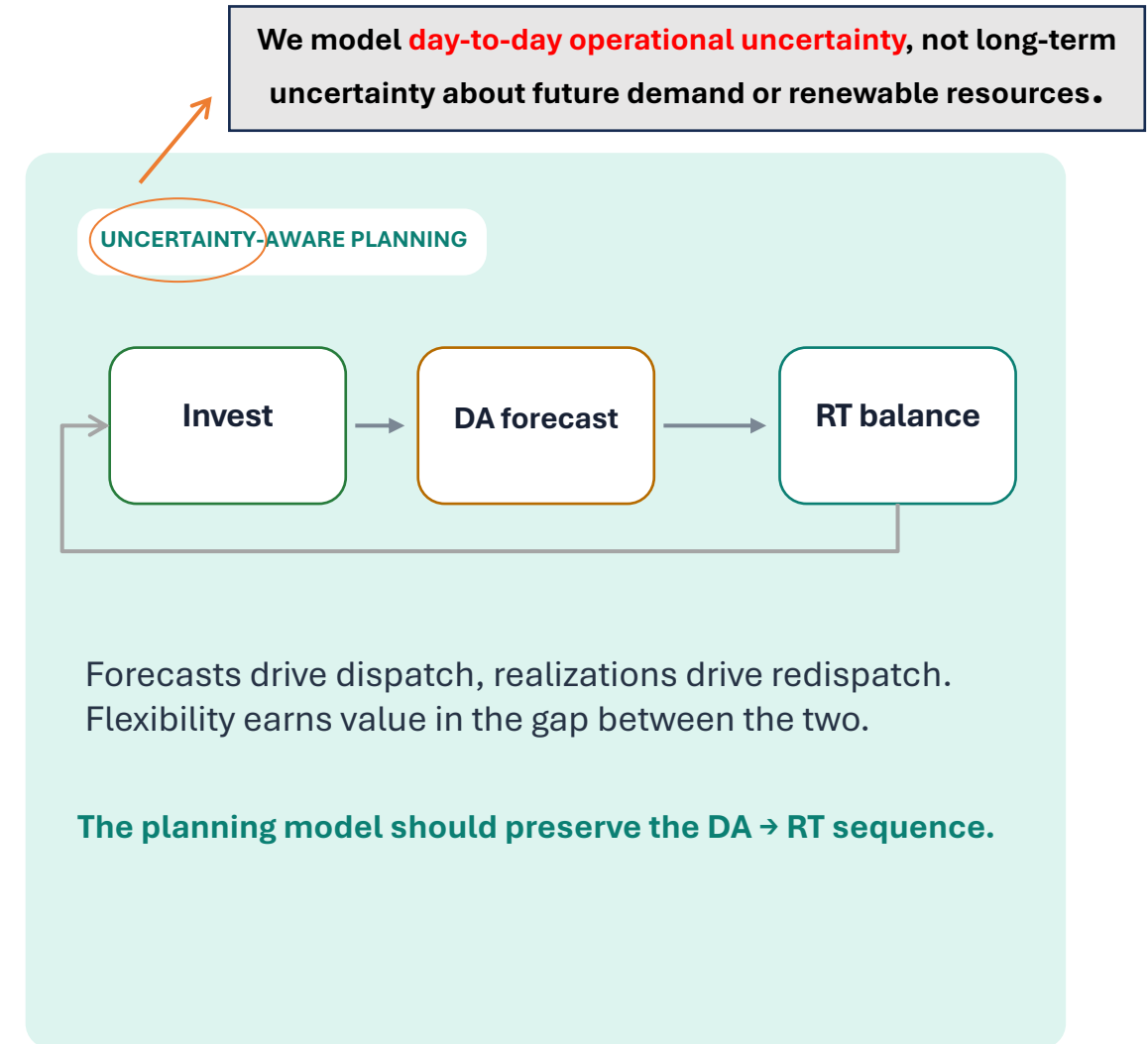
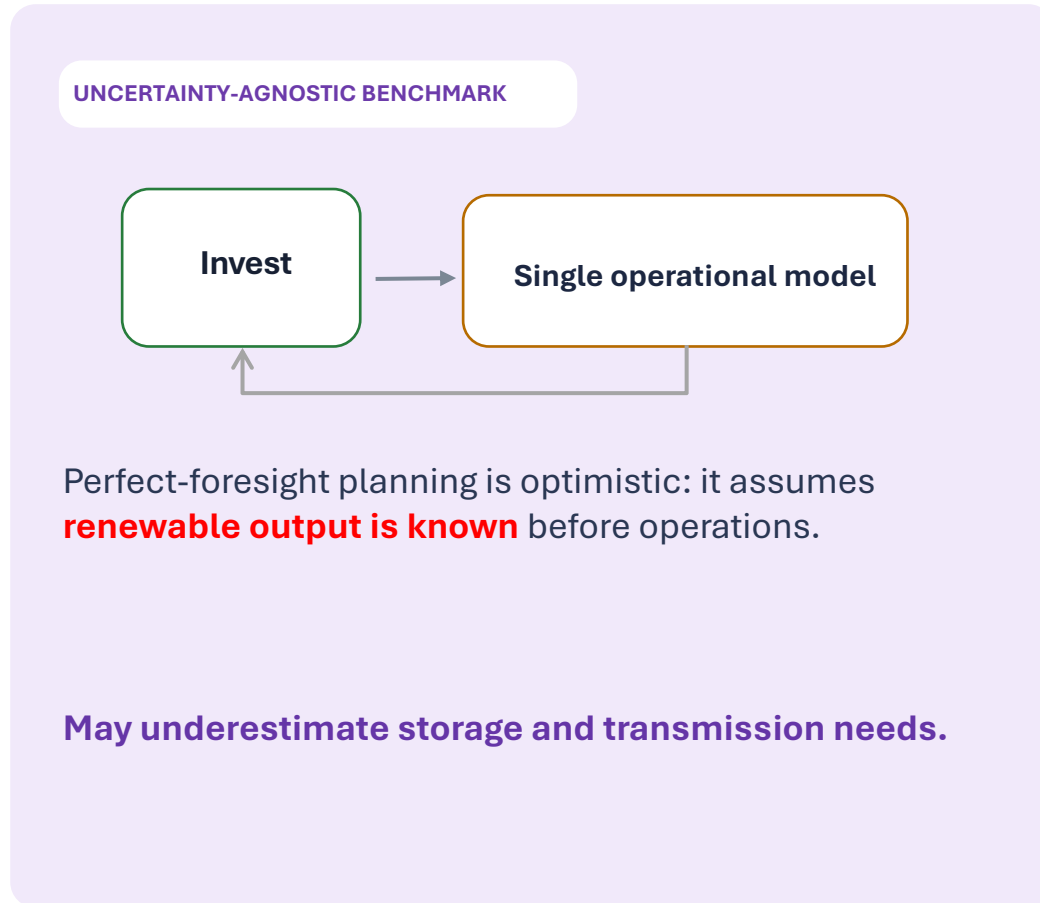


Perfect-foresight planning is optimistic: it assumes **renewable output is known** before operations.

May underestimate storage and transmission needs.

What standard expansion planning (uncertainty-agnostic model) misses

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A planner chooses investments while DA and RT operators solve their own problems

Planning objective + investment feasibility

$$\begin{aligned} \min \quad & J(\boldsymbol{\theta}) := \sum_{y \in [Y]} \left(\frac{1}{1+r} \right)^{\tau(y)} \left(C_y^{\text{inv}}(\boldsymbol{\theta}_y) + \sum_{d=1}^D \left(C_{d,y}^{\text{DA}}(\mathbf{x}_{d,y}^{\text{DA},*}(\boldsymbol{\theta}_y)) + C_{d,y}^{\text{RT}}(\mathbf{x}_{d,y}^{\text{RT},*}(\boldsymbol{\theta}_y, \mathbf{x}_{d,y}^{\text{DA},*}(\boldsymbol{\theta}_y))) \right) \right) \\ \text{s.t.} \quad & \underline{\boldsymbol{\theta}}_y \leq \boldsymbol{\theta}_y \leq \bar{\boldsymbol{\theta}}_y, \quad \forall y \in [Y], \\ & \boldsymbol{\theta}_{y-1} \leq \boldsymbol{\theta}_y, \quad \forall y \in [Y] \setminus \{1\}, \end{aligned}$$

Investment plan

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Investment plan

DA schedule → RT recourse

$$\begin{aligned} \underline{x}_{d,y}^{\text{DA},*}(\theta_y) &\in \left\{ \begin{array}{l} \arg \min_{\underline{x}_{d,y}^{\text{DA}}} C_{d,y}^{\text{DA}}(\underline{x}_{d,y}^{\text{DA}}) \\ \text{s.t.} \quad A_{d,y}^{\text{DA}}(\theta_y) \underline{x}_{d,y}^{\text{DA}} \leq b_{d,y}^{\text{DA}}(\theta_y, \bar{\xi}_{d,y}). \end{array} \right\} \quad \forall d \in [D], \forall y \in [Y], \\ \text{Day-ahead dispatch} \quad \underline{x}_{d,y}^{\text{RT},*}(\theta_y, \underline{x}_{d,y}^{\text{DA},*}(\theta_y)) &\in \left\{ \begin{array}{l} \arg \min_{\underline{x}_{d,y}^{\text{RT}}} C_{d,y}^{\text{RT}}(\underline{x}_{d,y}^{\text{RT}}) \quad \text{Forecast} \\ \text{s.t.} \quad A_{d,y}^{\text{RT}}(\theta_y, \underline{x}_{d,y}^{\text{DA},*}(\theta_y)) \underline{x}_{d,y}^{\text{RT}} \leq b_{d,y}^{\text{RT}}(\theta_y, \underline{x}_{d,y}^{\text{DA},*}(\theta_y), \bar{\xi}_{d,y}). \end{array} \right\} \\ \text{Real-time redispatch} \quad & \quad \quad \quad \forall d \in [D], \forall y \in [Y]. \quad \text{True value} \end{aligned}$$

A planner chooses investments while DA and RT operators solve their own problems

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Investment plan

Renewable-target constraint

Minimum renewable production share

$$\eta_y \left(\sum_{d \in [D]} \left(G^{\text{conv}}(\underline{x}_{d,y}^{\text{DA},*}, \underline{x}_{d,y}^{\text{RT},*}) + G^{\text{ren}}(\underline{x}_{d,y}^{\text{RT},*}) \right) \right) \leq \sum_{d \in [D]} G^{\text{ren}}(\underline{x}_{d,y}^{\text{RT},*}), \quad \forall y \in [Y],$$

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Day-ahead dispatch

Real-time redispatch

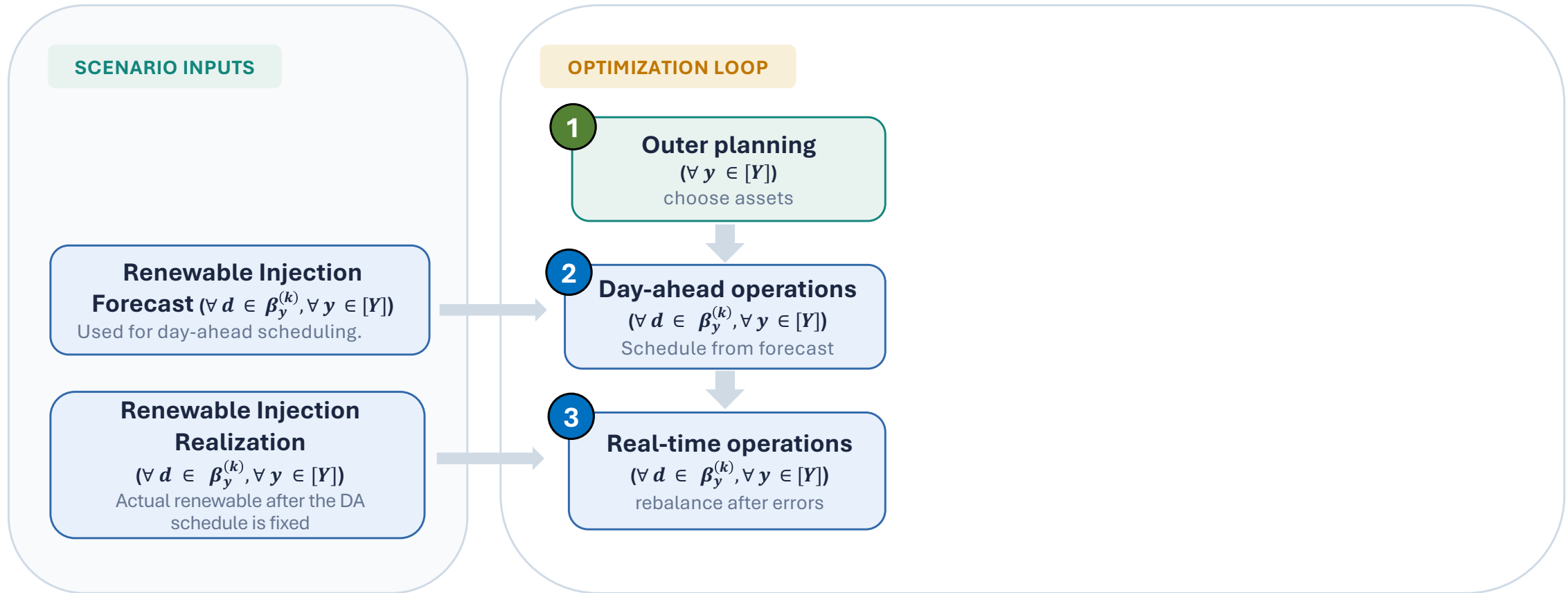
This is a **multi-level** planning problem: the outer planner cannot directly choose DA/RT operating actions.

Projected stochastic gradient descent:

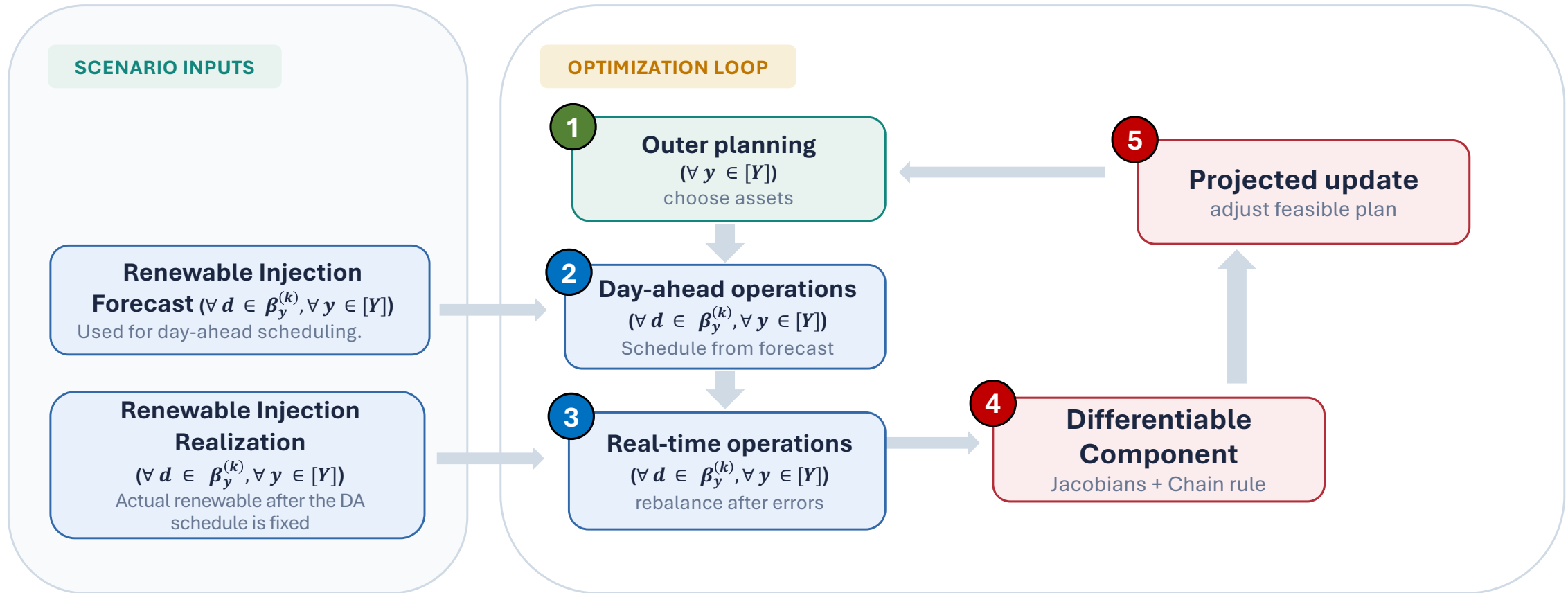
SCENARIO INPUTS

$$(\forall d \in \beta_y^{(k)}, \forall y \in [Y])$$

Projected stochastic gradient descent:



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Planning objective + investment feasibility

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Renewable target

Planning objective + investment feasibility

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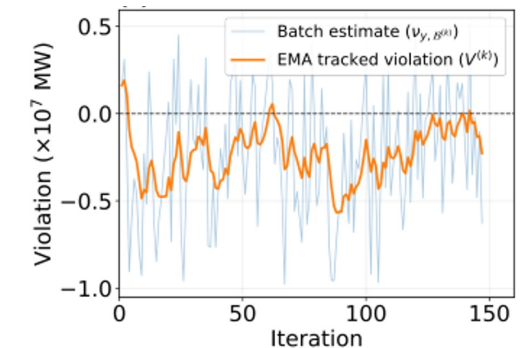
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Key difficulty: the constraint depends on **optimized operating decisions**, not only capacity expansion decision variables.

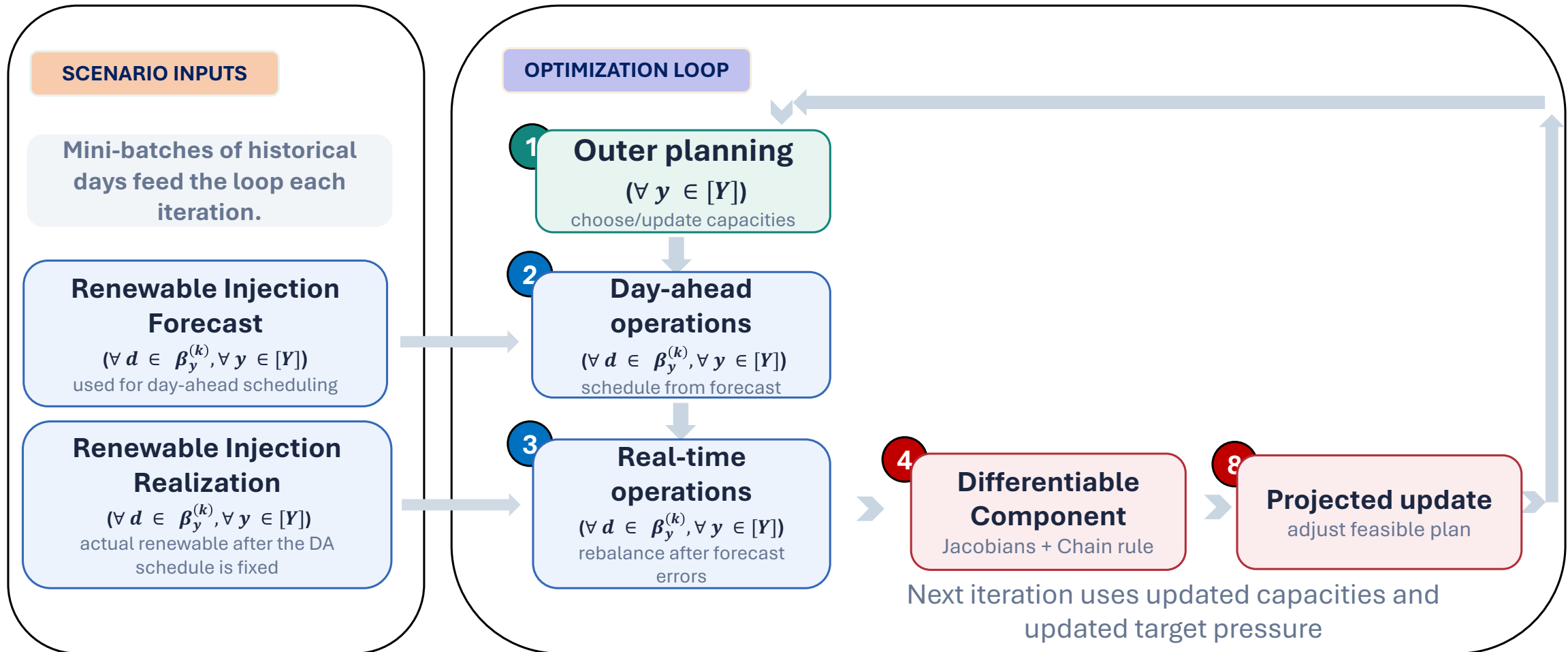
As a result, the projection can not be applied to this constraint.

We define **In-Batch shortfall**:
 $v > 0$ means a renewable shortfall;
 $v \leq 0$ means the target is satisfied.



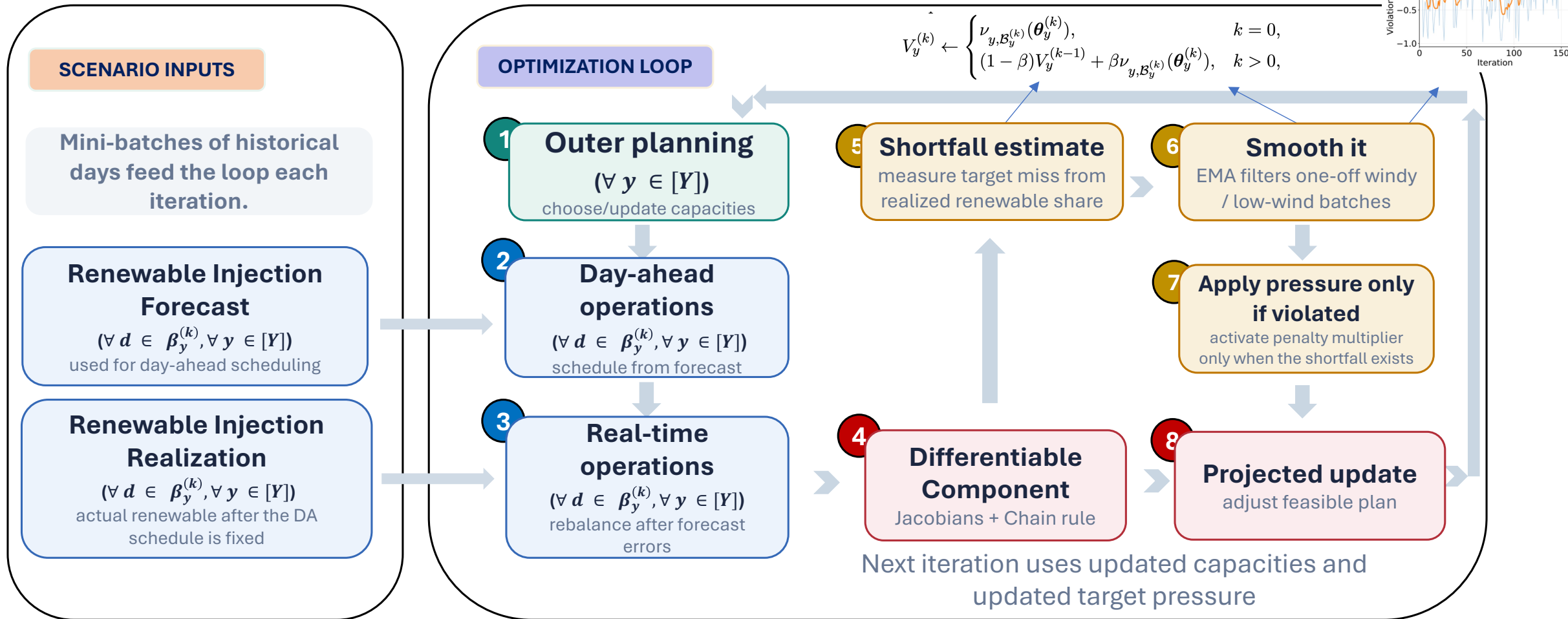
How the model responds to renewable-target violations

Starting from the basic DA → RT loop, we add a second signal that tracks and stabilizes renewable-target violations over iterations.

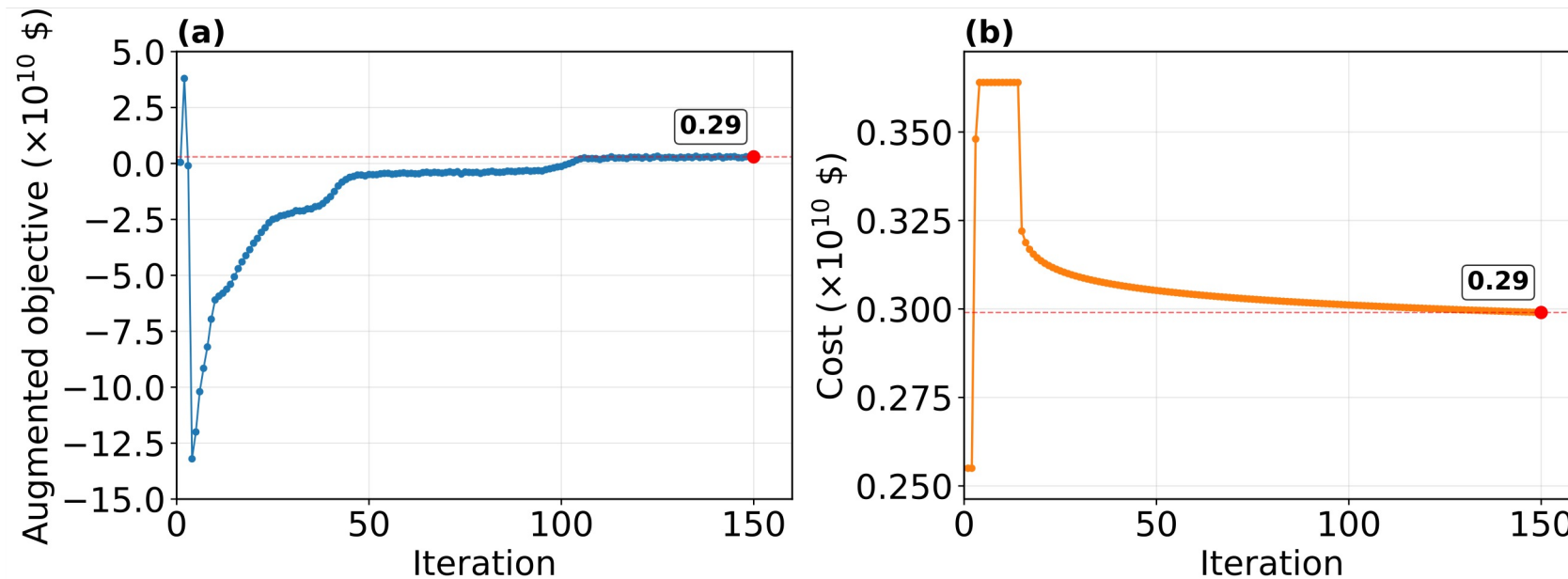


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Convergence behavior



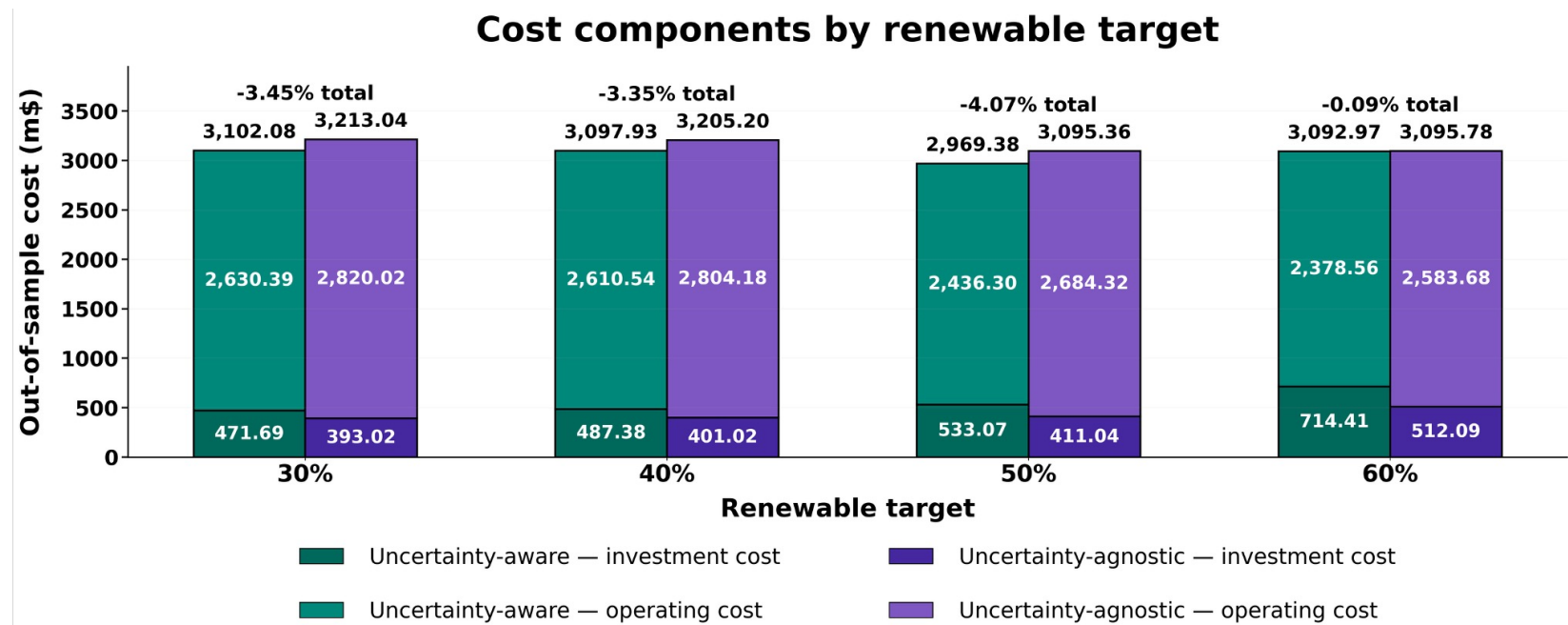
a Projected SGD convergence with the primal–dual method
Renewable target: 50% | 5 sampled days per iterations

b Planning objective without the renewable-target penalty
Renewable target: 50% | 5 sampled days per iterations

Augmented objective and cost settle to the same final level once the target is satisfied.

Impact of forecast errors on expansion planning

Uncertainty-aware planning buys flexibility: **higher investment cost**, **lower dispatch cost**, and **better target achievement**.



Renewable Target		30%	40%	50%	60%
Realized Renewable	U_AW	30%	40%	50%	60%
	U-AG	30%	39%	45%	52%

Forecast error is not just an operational problem. It changes the best infrastructure plan.



Please scan to read our paper!

**Feel free to reach out to me at:
Email : Mehnoush.Ghazanfariharandi@rutgers.edu**

Thank you!